ECE 3355: Electronics

MidTerm Exam 2

March 29, 2025

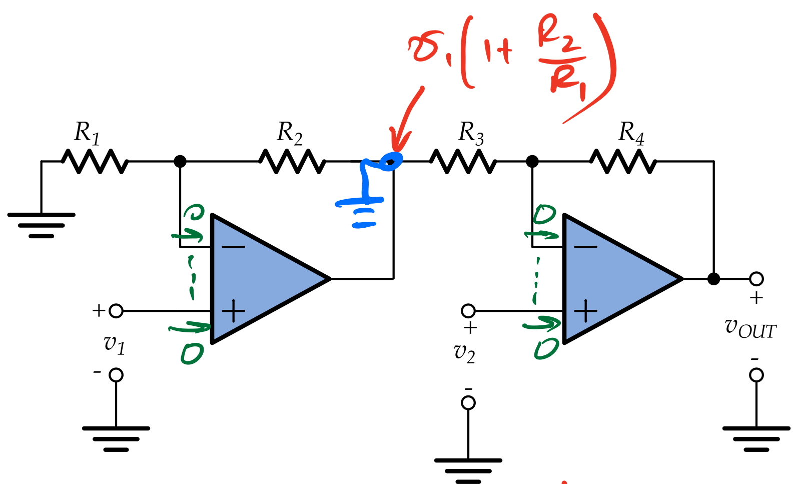
Do *not* open the exam until instructed to do so. Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page. This is a closed-book/notes exam and you *may* use a calculator. You must present your work in a clear and concise manner to receive credit for a correct solution. **You will have 1 hours 15 min to inish the exam.**

Student's Name: ______________________________

Question	Points	Score
1	30	
2	40	
3	30	
Total:	100	

1. (30 points) For the following difference amplifier, $R_1 = R_4 = 9\text{ k}\Omega$ and $R_2 = R_3 = 1\text{ k}\Omega$. Assume ideal op-amps. Find
   1. A_d,
   2. A_{cm} when $R_3 = 1.03\text{ k}\Omega$,
   3. the CMRR, and
   4. the two input resistances (as seen at the two input terminals).

   Hints: Use superposition. For calculating the CMRR, use the A_d from the first part – there is no need to attempt to recalculate it for the circuit with the new resistance. *Important:* show your work in a way that allows me to see how you solved this problem to receive credit.



$\boxed{v_1 \neq 0,\ v_2 = 0} \Rightarrow v_{o_1} = -\frac{R_4}{R_3}\left(1+\frac{R_2}{R_1}\right)v_1$

$\boxed{v_1 = 0,\ v_2 \neq 0} \Rightarrow v_{o_2} = \left(1+\frac{R_4}{R_3}\right)v_2$

$$v_o = \underbrace{\left(1+\frac{R_4}{R_3}\right)}_{A_d = 10} v_2 - \underbrace{\left(1+\frac{R_2}{R_1}\right)\frac{R_4}{R_3}}_{A_d = 1} \cdot v_1$$

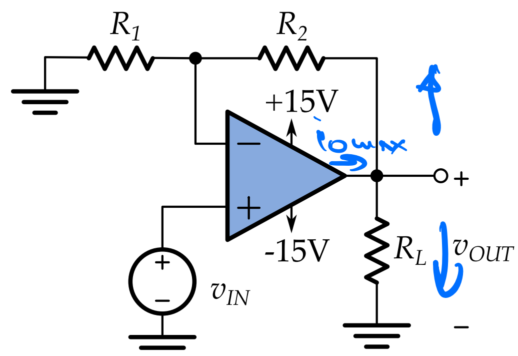
$$1+\frac{R_4}{R_3} = \left(1+\frac{R_2}{R_1}\right)\frac{R_4}{R_3} = \frac{R_4}{R_3} + \frac{R_2}{R_1}\frac{R_4}{R_3} \Rightarrow \frac{R_2}{R_1}\frac{R_4}{R_3} = 1$$

← condition for the circuit to work as diff. ampl.

$v_1 = v_2 = v_{cm}$ $\quad v_o = \underbrace{\left[1+\frac{R_4}{R_3} - \left(1+\frac{R_2}{R_1}\right)\frac{R_4}{R_3}\right]}_{A_{cm} = 0{,}029} v_{cm}$

$\text{CMRR} = 20\log\frac{10}{0{,}029} = 50\text{ dB}$

2. (40 points) Determine if the op-amp is in saturation (either current or voltage saturation) – you must show work that indicates that you have tested for each case. What is the voltage v_{OUT}, assuming an ideal op-amp? Here, $R_1 = 1$ kΩ, $R_2 = 9$ kΩ, $R_L = 2$ kΩ, $v_{IN} = 2$ V, and $i_{max} = 10$ mA.



(1) if $V_{IN} = 2 \Rightarrow V_{out} = 20$ ← voltage saturation

(2) $V_{out} = 15$ V ← need to check for current saturation

* no virtual short !!!

$$i_o = \frac{v_{out}}{R_L} + \frac{v_{out}}{R_1 + R_2}$$

$$i_o = \frac{15}{2k} + \frac{15}{10k} = 7.5m + 1.5m = 9m < 10mA$$

no current saturation

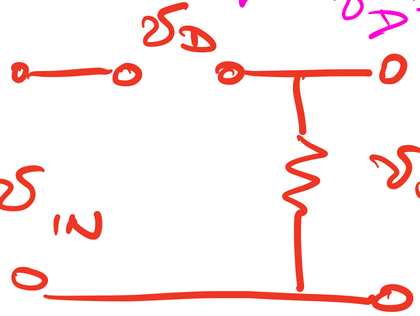
$$\underline{v_{out} = 15 \text{ V}}$$

3. (30 points) For the circuit below, assume $V_{D0} = 0.5$ volts, $R_1 = r_D$, $R_2 = 2r_Z$, $r_D = r_Z$, and $V_{Z0} = 2$ volts.

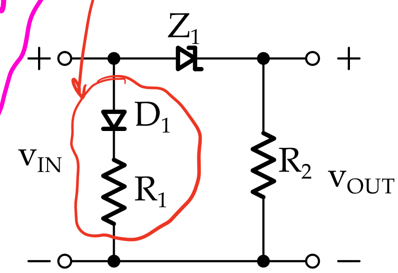
small v_{IN}

$v_D = v_{IN}$

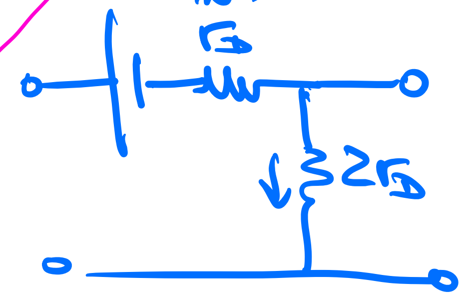
$v_o = 0$



does not contribute to v_{out}



$v_{IN} > 0.5$



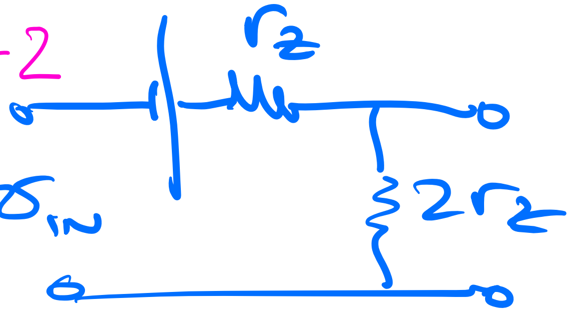
$$I = \frac{v_{IN} - 0.5}{3r_D}$$

$$V_o = 2r_D I = \frac{v_{IN} - 0.5}{3} \cdot 2$$

1. Write an expression for v_{OUT} in terms of v_{IN}.

$$-2 < v_{IN} < 0.5 \; : \; v_o = 0$$

$v_{IN} \leq -2$



$$I = \frac{v_{IN} + 2}{3r_Z}$$

$$v_o = 2r_Z \cdot I = \frac{v_{IN} + 2}{3} \cdot 2$$

2. Sketch v_{OUT} vs. v_{IN} for -5 volts $< v_{IN} < 5$ volts on the graph provided. Be sure to label the graph properly.